



Organic Chemicals of Life

Carbohydrates

Lipids

Proteins

Nucleic Acids

IJONAI Emmanuel

CARBOHYDRATES

- These comprise of a large group of organic compounds which contain **C**, **H** and **O**.
- They have a general formula **$C_x(H_2O)_y$** though some do not conform to it e.g. deoxyribose **$C_5H_{10}O_4$** .

Main functions of carbohydrates

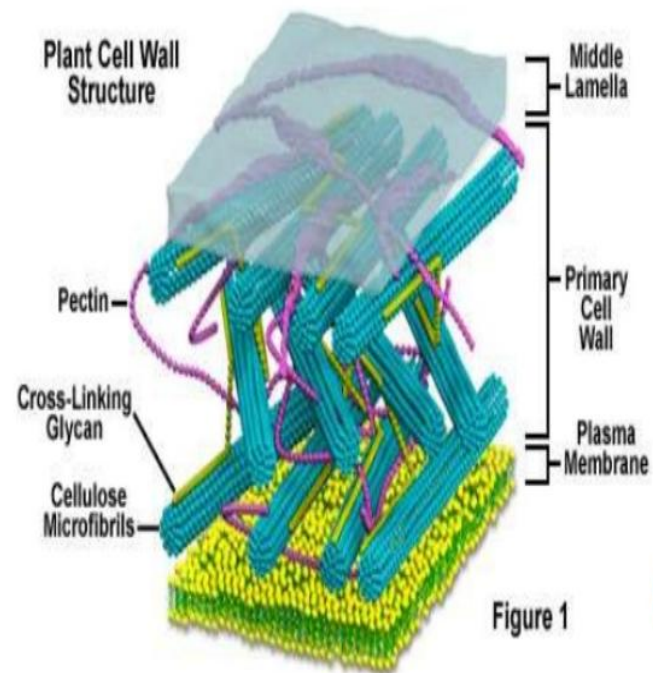
1. They are a primary source of **energy** being oxidized in the body to release energy.



Main functions of carbohydrates

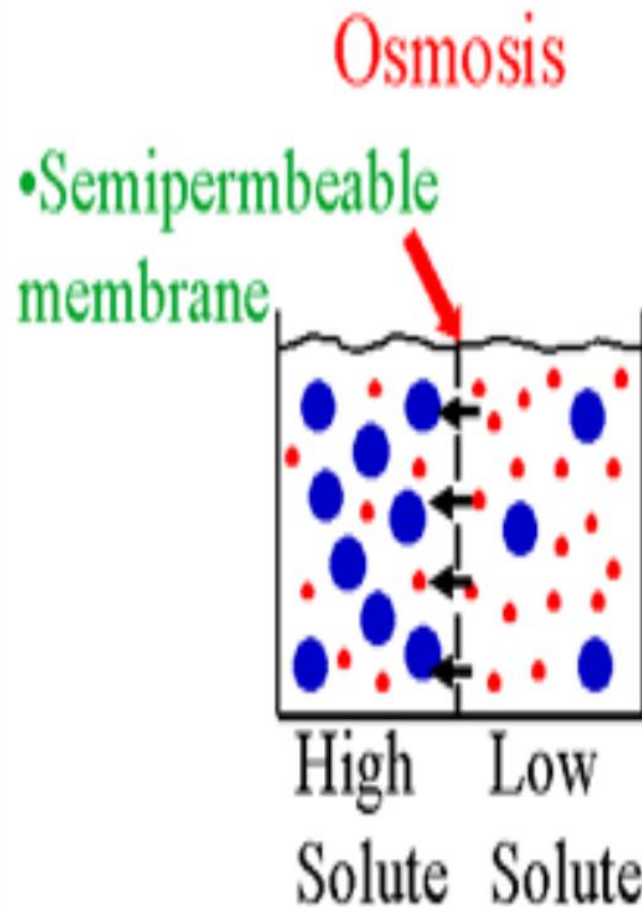
2. They are structural ***components of cells*** e.g. cellulose making up the cell wall.

PLANT CELL WALL STRUCTURE



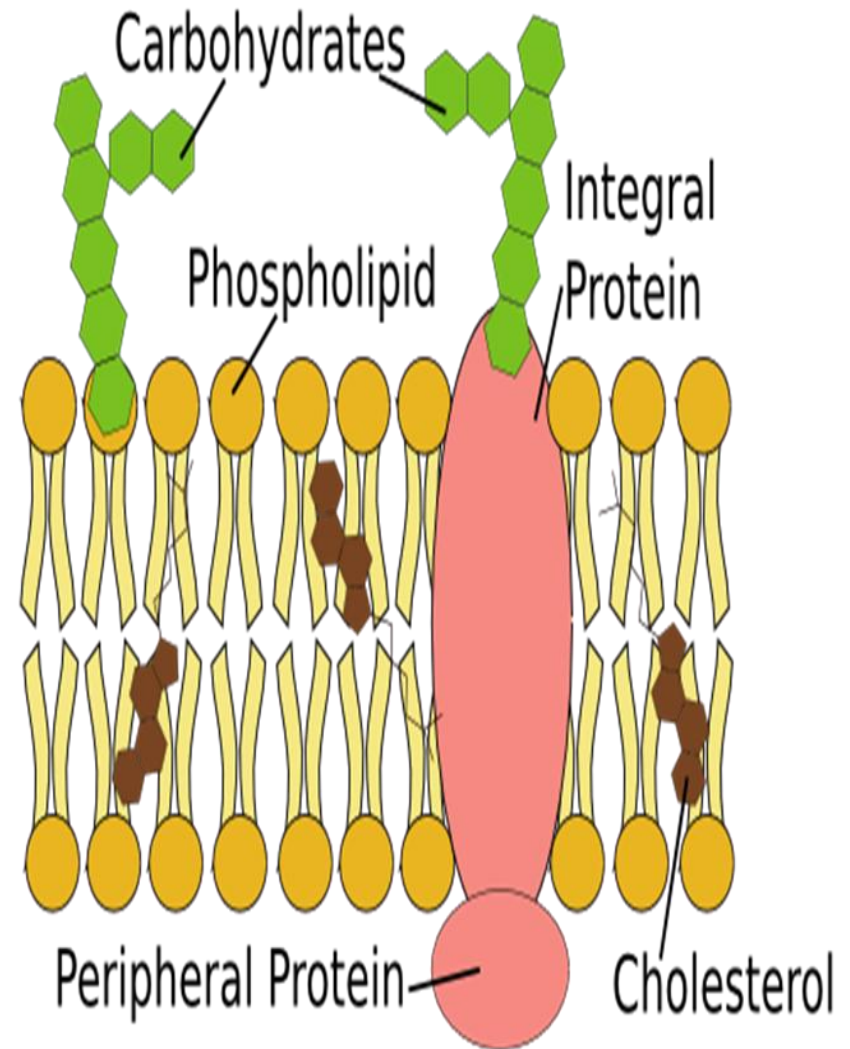
Main functions of carbohydrates

3. They are determinants of **osmotic potential** of body fluids therefore maintain blood pressure.



Main functions of carbohydrates

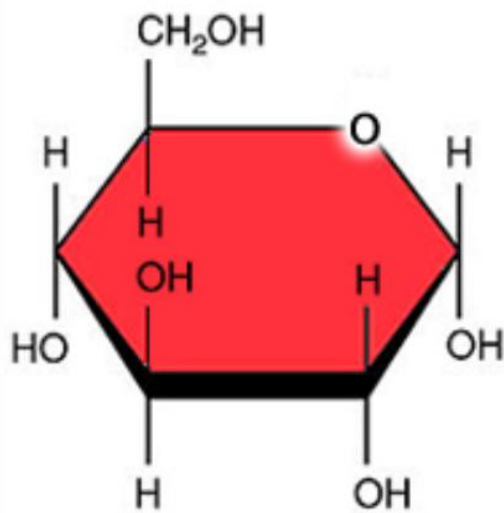
4. They are recognized units on the surface of body cells, i.e. they are component ***structures of the surface cell membranes*** recognized by antibodies.



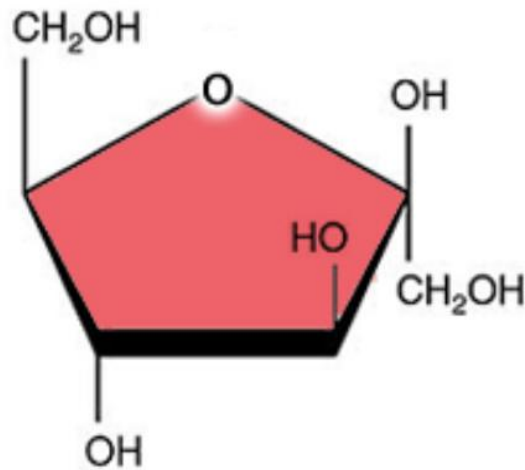
TYPES OF CARBOHYDRATES

- **Monosaccharides** (single unit sugars)
- **Disaccharides** (double unit sugars)
- **Polysaccharides** (several unit sugars)

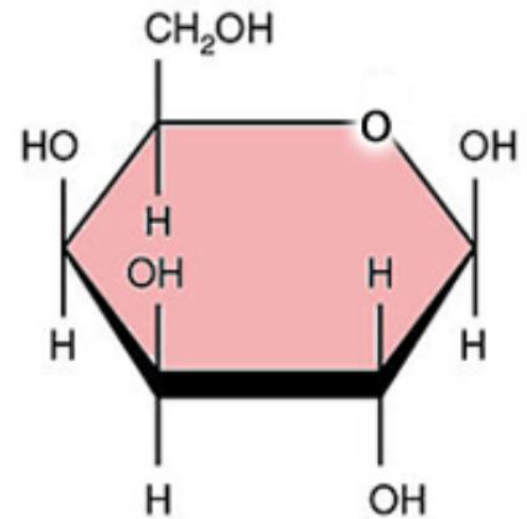
Monosaccharides



Glucose



Fructose



Galactose

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MONOSACCHARIDES

Monosaccharides (**mono**=one, **saccharide**= sugar) are substances consisting of one molecule of sugar.

- They are also known as simple sugars.

Properties of Monosaccharides

a) They
have a
sweet
taste



Properties of Monosaccharides

b) They
dissolve
in water



Properties of Monosaccharides

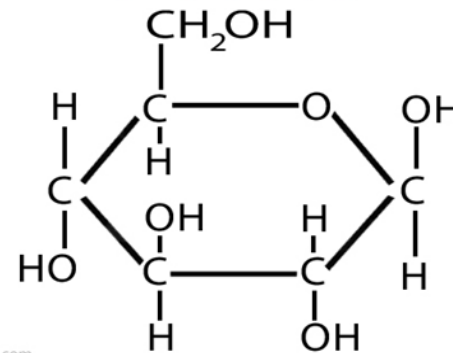
c) They
form
crystals



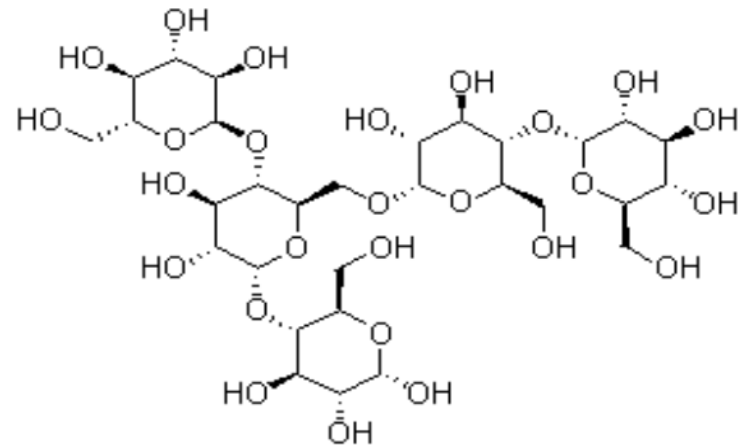
Properties of Monosaccharides

d) They have
a **low**
molecular
mass

Glucose

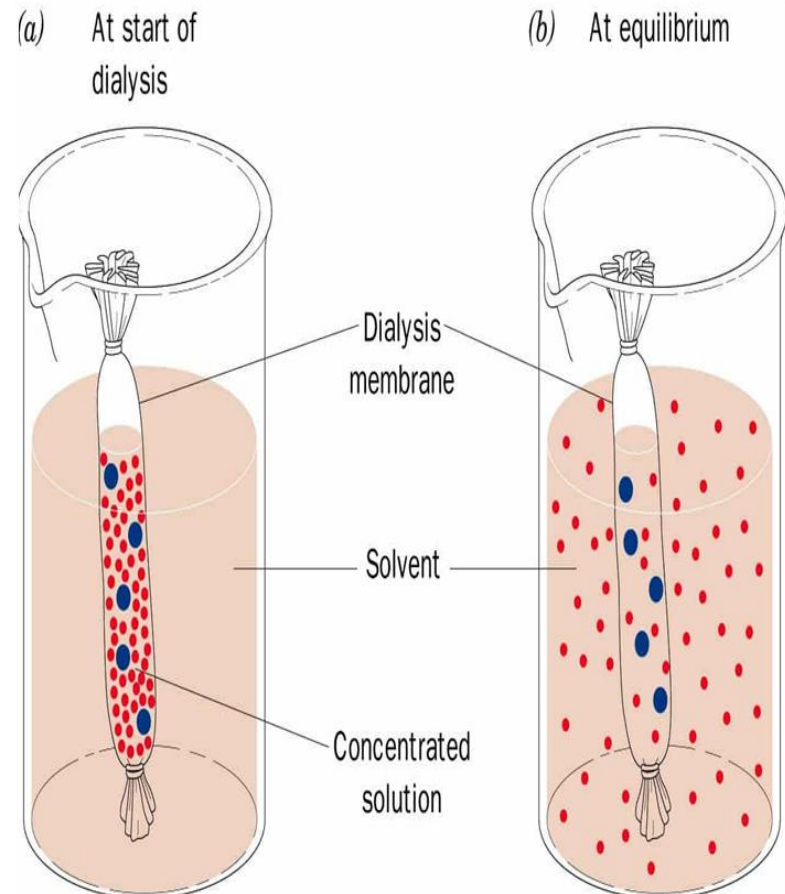


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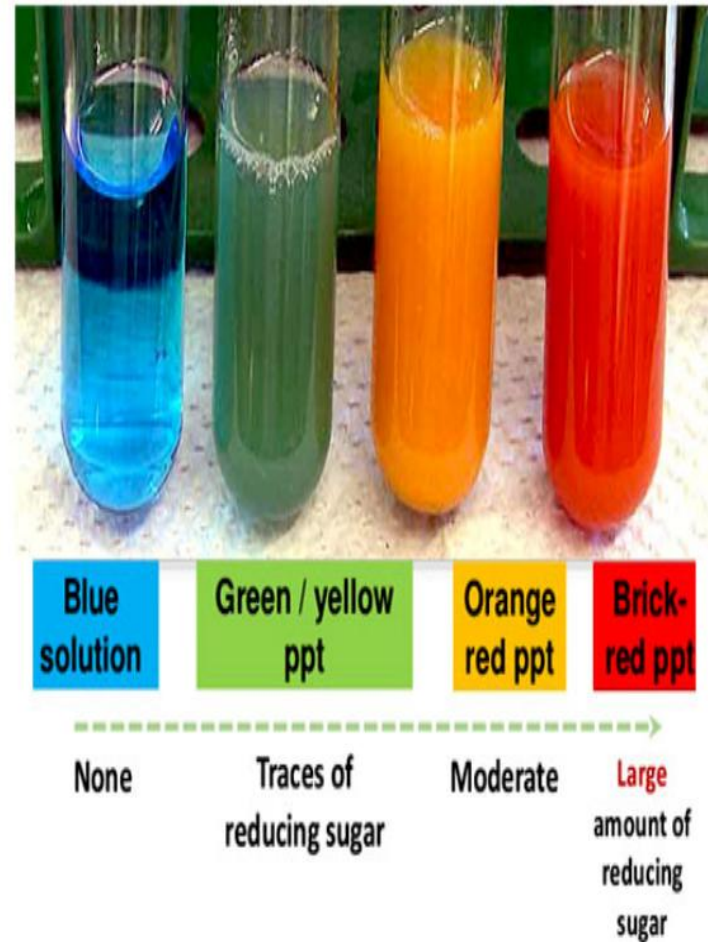
Properties of Monosaccharides

e) *Can pass through a selectively permeable membrane.*



Properties of Monosaccharides

f) They change the colour of **Benedict's** solution from blue to orange when boiled with the solution thus they are known as ***reducing sugars***.



Monosaccharides

- ❖ Monosaccharides are named using a suffix '**ose**'.
- ❖ They contain either an **aldehyde** group (**CHO**) and are called aldoses or they contain **a ketone** group (**C=O**) and are called ketones.
- ❖ Monosaccharides have a general formula **(CH₂O)_n**

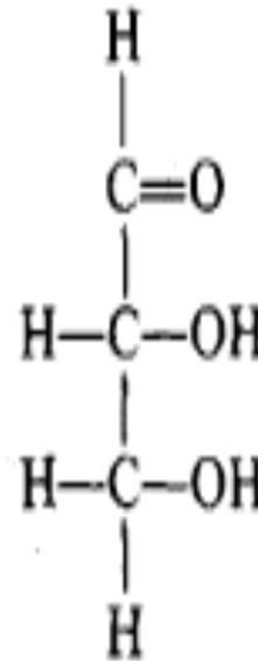
Classification of Monosaccharides

Basing on number of carbon atoms

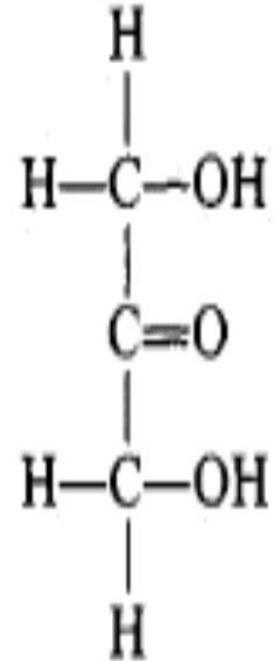
1. **n=3** (*triose sugar*)
2. **n=5** (*pentose sugar*)
3. **n=6** (*hexose sugar*)
4. **n=7** (*heptose*), etc

Triose sugar

1. A **triose** is a monosaccharide, or simple **sugar**, containing three carbon atoms.



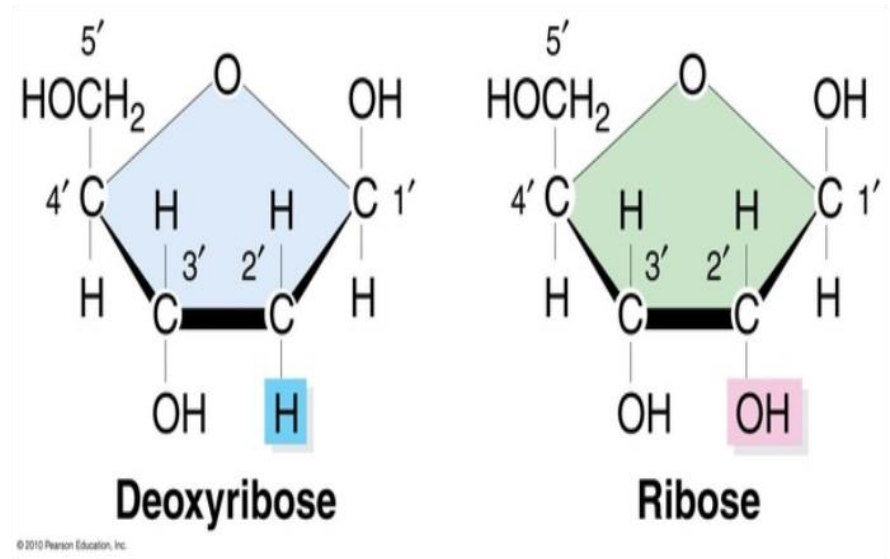
D = glyceraldehyde
(aldose)



Dihydroxyacetone
(ketose)

Pentose sugars

2. The **sugars** found in nucleic acids are **pentose sugars**; a **pentose sugar** has five carbon atoms



Hexose sugars

3. a hexose

is a
monosaccharide
(simple
sugar) with
six carbon
atoms.

